

Objective:

This project analyzed simulated auto-insurance policy data to determine how expected insurance costs vary by age group, territory, and vehicle type.

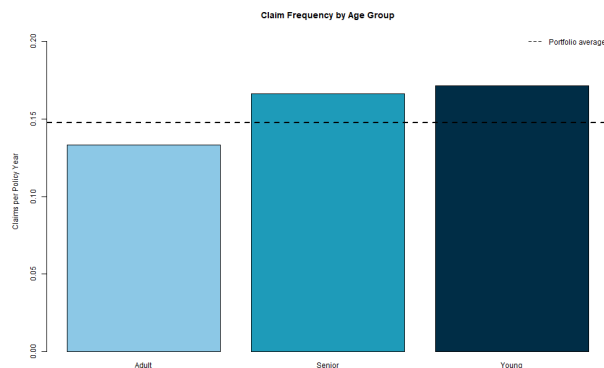
Data and methods:

This project uses simulated data created in R. The dataset contains 3,000 simulated auto-insurance policies. I used R to generate policy characteristics, claim amounts, and claim counts based on actuarial-style assumptions. I then calculated claim frequency, claim severity, pure premium, and indicated premium for each risk group.

Limitation:

One limitation is that the data was simulated rather than collected from a real insurer. Insurers should also consider additional variables such as driving history, annual miles, prior claims, credit score, and more when setting premium prices.

	rating_variable	group	exposure	claims	losses	claim_frequency	claim_severity	pure_premium	indicated_premium
1	age_group	Adult	1538.31	205	841236.7	0.1333	4103.59	546.86	781.23
2	age_group	Senior	384.86	64	265376.8	0.1663	4146.51	689.54	985.06
3	age_group	Young	623.53	107	482689.3	0.1716	4511.12	774.12	1105.89
4	territory	Rural	494.34	51	176629.5	0.1032	3463.32	357.30	510.43
5	territory	Suburban	1143.63	143	586154.7	0.1250	4098.98	512.54	732.20
6	territory	Urban	908.73	182	826518.7	0.2003	4541.31	909.53	1299.33
7	vehicle_type	Sedan	1391.11	186	725649.7	0.1337	3901.34	521.63	745.19
8	vehicle_type	SUV	1155.59	190	863653.1	0.1644	4545.54	747.37	1067.67



Key findings:

- Young adults had the highest indicated premium at \$1,105.89, compared with \$985.06 for seniors and \$781.23 for adults. Young adults also had the highest claim frequency at 0.1716 claims per policy year, compared with 0.1663 for seniors and 0.1333 for adults. In the simulation, young drivers were assumed to have 45% higher expected claim frequency than adults, while seniors were assumed to have 20% higher frequency. Adults had the lowest indicated premium because they were used as the baseline group. These assumptions reflect broad auto-insurance patterns. Young drivers are statistically more accident-prone, while some older drivers may experience age-related changes that impact their driving ability.

- Urban territories had a higher expected claim frequency and claim severity than suburban and rural areas. The simulation assumed urban areas had a higher risk to reflect real-world factors that may affect auto insurance losses in dense urban areas, such as heavy traffic and higher rates of car theft. Because the urban policies had a higher expected claim frequency and severity than the other territories, they also had the highest pure and indicated premium.
- SUVs had a higher pure premium than sedans. This is because SUVs have both higher claim frequency (0.1644 vs 0.1337) and higher claim severity (\$4545.54 vs \$3901.34). The simulation assumed SUVs have a 10% higher expected claim frequency and 25% higher expected claim severity than sedans, reflecting possible differences in repair costs and potential accident damage.

Conclusion:

Based on my analysis, urban territories had the highest expected claim costs and indicated premium. Young adults and SUV owners also had higher expected loss costs than their comparison groups. An insurer should consider charging higher premiums for these characteristics, while also using additional information such as driving records before making final pricing decisions.

